Calculus 1 Problem Templates for Dynamic Generation

Master Reference Sheet for Generator Patterns

Calculus Practice System (v.2026.01)

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Instructions for Professor: This document collects representative problem archetypes across all 7 core categories in the Calculus 1 practice hub. Please write your step-by-step gold-standard solution and LaTeX layout inside the `solutionbox` for each problem type. Once completed, the system will use your exact structure as the universal dynamic template for all randomized problem instances.

Contents

1 Limits of Functions

1.1 Type 1.1: Factoring (0/0 Quadratic / Difference of Squares)

Problem 1.1: Limit by Factoring

Evaluate the limit:

$$\lim_{x \to 2} \frac{x^2 - 4}{x - 2}$$

Gold-Standard Solution (Write your master solution template here)

Step 1. Check the indeterminate form by direct substitution:

$$\frac{2^2 - 4}{2 - 2} = \frac{0}{0} \quad \text{(Indeterminate Form).}$$

Step 2. Factor the numerator using the difference of squares identity $a^2 - b^2 = (a-b)(a+b)$:

$$x^2 - 4 = (x - 2)(x + 2).$$

Step 3. Simplify the rational function by cancelling the common non-zero factor $(x - 2)$ for $x \neq 2$:

$$\frac{x^2 - 4}{x - 2} = \frac{(x - 2)(x + 2)}{x - 2} = x + 2.$$

Step 4. Evaluate the limit by direct substitution:

$$\lim_{x \to 2} \frac{x^2 - 4}{x - 2} = \lim_{x \to 2}(x + 2) = 2 + 2 = 4.$$

1.2 Type 1.2: Square Root Conjugate (0/0)

Problem 1.2: Limit by Conjugate (Square Root)

Evaluate the limit:

$$\lim_{x \to 0} \frac{\sqrt{x + 4} - 2}{x}$$

Gold-Standard Solution (Write your master solution template here)

Step 1. Check the indeterminate form by direct substitution:

$$\frac{\sqrt{0 + 4} - 2}{0} = \frac{2 - 2}{0} = \frac{0}{0} \quad \text{(Indeterminate Form).}$$

Step 2. Multiply the numerator and denominator by the conjugate $(\sqrt{x + 4} + 2)$:

$$\frac{\sqrt{x + 4} - 2}{x} \cdot \frac{\sqrt{x + 4} + 2}{\sqrt{x + 4} + 2} = \frac{(\sqrt{x + 4})^2 - 2^2}{x(\sqrt{x + 4} + 2)}.$$

Step 3. Simplify the expression:

$$\frac{(x + 4) - 4}{x(\sqrt{x + 4} + 2)} = \frac{x}{x(\sqrt{x + 4} + 2)} = \frac{1}{\sqrt{x + 4} + 2}.$$

Step 4. Evaluate the limit by direct substitution $x = 0$:

$$\lim_{x\to 0} \frac{\sqrt{x+4}-2}{x} = \lim_{x\to 0} \frac{1}{\sqrt{x+4}+2} = \frac{1}{\sqrt{0+4}+2} = \frac{1}{2+2} = \frac{1}{4}.$$

1.3 Type 1.3: Cube Root Conjugate / Difference of Cubes (0/0)

Problem 1.3: Limit by Cube Root Conjugate

Evaluate the limit:

$$\lim_{x\to 0} \frac{\sqrt[3]{x+8}-2}{x}$$

Gold-Standard Solution (Write your master solution template here)

Step 1. Check the indeterminate form by direct substitution:

$$\frac{\sqrt[3]{0+8}-2}{0} = \frac{2-2}{0} = \frac{0}{0} \quad \text{(Indeterminate Form)}.$$

Step 2. Use the identity $(a-b)(a^2+ab+b^2) = a^3-b^3$. Multiply numerator and denominator by $(\sqrt[3]{x+8})^2 + 2\sqrt[3]{x+8} + 4$:

$$\frac{\sqrt[3]{x+8}-2}{x} \cdot \frac{(\sqrt[3]{x+8})^2 + 2\sqrt[3]{x+8}+4}{(\sqrt[3]{x+8})^2 + 2\sqrt[3]{x+8}+4} = \frac{(x+8)-8}{x\left((\sqrt[3]{x+8})^2 + 2\sqrt[3]{x+8}+4\right)}.$$

Step 3. Simplify the expression:

$$\frac{x}{x\left((\sqrt[3]{x+8})^2 + 2\sqrt[3]{x+8}+4\right)} = \frac{1}{(\sqrt[3]{x+8})^2 + 2\sqrt[3]{x+8}+4}.$$

Step 4. Evaluate the limit by direct substitution $x = 0$:

$$\lim_{x\to 0} \frac{1}{(\sqrt[3]{8})^2 + 2\sqrt[3]{8}+4} = \frac{1}{4+4+4} = \frac{1}{12}.$$

1.4 Type 1.4: Complex Fraction (0/0)

Problem 1.4: Limit of Complex Fractions

Evaluate the limit:

$$\lim_{x\to 0} \frac{\frac{1}{x+3}-\frac{1}{3}}{x}$$

Gold-Standard Solution (Write your master solution template here)

Step 1. Check the indeterminate form by direct substitution:

$$\frac{\frac{1}{0+3}-\frac{1}{3}}{0} = \frac{0}{0} \quad \text{(Indeterminate Form)}.$$

Step 2. Combine the terms in the numerator with a common denominator $3(x+3)$:

$$\frac{1}{x+3} - \frac{1}{3} = \frac{3-(x+3)}{3(x+3)} = \frac{-x}{3(x+3)}.$$

Step 3. Simplify the complex fraction:

$$\frac{\frac{-x}{3(x+3)}}{x} = \frac{-x}{3x(x+3)} = -\frac{1}{3(x+3)}.$$

Step 4. Evaluate the limit by direct substitution $x=0$:

$$\lim_{x\to 0} -\frac{1}{3(x+3)} = -\frac{1}{3(0+3)} = -\frac{1}{9}.$$

1.5 Type 1.5: Double Conjugate (Numerator & Denominator)

Problem 1.5: Double Conjugate Limit

Evaluate the limit:

$$\lim_{x\to 0} \frac{\sqrt{x+4}-2}{\sqrt{x+9}-3}$$

Gold-Standard Solution (Write your master solution template here)

Step 1. Check the indeterminate form by direct substitution:

$$\frac{\sqrt{4}-2}{\sqrt{9}-3} = \frac{2-2}{3-3} = \frac{0}{0} \quad \text{(Indeterminate Form)}.$$

Step 2. Multiply by both conjugates $(\sqrt{x+4}+2)$ and $(\sqrt{x+9}+3)$:

$$\frac{\sqrt{x+4}-2}{\sqrt{x+9}-3} \cdot \frac{\sqrt{x+4}+2}{\sqrt{x+4}+2} \cdot \frac{\sqrt{x+9}+3}{\sqrt{x+9}+3} = \frac{((x+4)-4)(\sqrt{x+9}+3)}{((x+9)-9)(\sqrt{x+4}+2)}.$$

Step 3. Simplify and cancel common factor x:

$$\frac{x(\sqrt{x+9}+3)}{x(\sqrt{x+4}+2)} = \frac{\sqrt{x+9}+3}{\sqrt{x+4}+2}.$$

Step 4. Evaluate the limit by direct substitution $x=0$:

$$\lim_{x\to 0} \frac{\sqrt{x+9}+3}{\sqrt{x+4}+2} = \frac{\sqrt{9}+3}{\sqrt{4}+2} = \frac{3+3}{2+2} = \frac{6}{4} = \frac{3}{2}.$$

1.6 Type 1.6: Basic Trigonometric Limits ($\frac{\sin kx}{x}$, $\frac{\tan kx}{\sin mx}$)

Problem 1.6: Trigonometric Limit

Evaluate the limit:

$$\lim_{x\to 0} \frac{\tan(3x)}{\sin(5x)}$$

Gold-Standard Solution (Write your master solution template here)

Step 1. Rules and standard trigonometric limits:

$$\lim_{u\to 0} \frac{\sin(u)}{u} = 1$$
$$\lim_{u\to 0} \frac{\tan(u)}{u} = 1$$

Step 2. Multiply and divide by $3x$ and $5x$ to obtain standard limit forms:

$$\frac{\tan(3x)}{\sin(5x)} = \frac{\frac{\tan(3x)}{3x}\cdot 3x}{\frac{\sin(5x)}{5x}\cdot 5x} = \frac{3}{5}\cdot\frac{\frac{\tan(3x)}{3x}}{\frac{\sin(5x)}{5x}}.$$

Step 3. Evaluate the limit as $x \to 0$:

$$\lim_{x\to 0} \frac{\tan(3x)}{\sin(5x)} = \frac{3}{5}\cdot\frac{\lim_{x\to 0}\frac{\tan(3x)}{3x}}{\lim_{x\to 0}\frac{\sin(5x)}{5x}} = \frac{3}{5}\cdot\frac{1}{1} = \frac{3}{5}.$$

1.7 Type 1.7: Limits at Infinity (∞/∞ & $\infty - \infty$)

Problem 1.7: Limit at Infinity

Evaluate the limit:

$$\lim_{x\to\infty} \left(\sqrt{x^2+4x} - x\right)$$

Gold-Standard Solution (Write your master solution template here)

Step 1. Check the indeterminate form: $\infty - \infty$.

Step 2. Multiply and divide by the conjugate $(\sqrt{x^2+4x}+x)$:

$$\frac{(\sqrt{x^2+4x}-x)(\sqrt{x^2+4x}+x)}{\sqrt{x^2+4x}+x} = \frac{(x^2+4x)-x^2}{\sqrt{x^2+4x}+x} = \frac{4x}{\sqrt{x^2+4x}+x}.$$

Step 3. Divide numerator and denominator by x (noting $\sqrt{x^2} = x$ for $x > 0$):

$$\frac{\frac{4x}{x}}{\sqrt{\frac{x^2+4x}{x^2}}+\frac{x}{x}} = \frac{4}{\sqrt{1+\frac{4}{x}}+1}.$$

Step 4. Evaluate the limit as $x \to \infty$, since $\lim_{x\to\infty} \frac{4}{x} = 0$:

$$\lim_{x\to\infty} \frac{4}{\sqrt{1+\frac{4}{x}}+1} = \frac{4}{\sqrt{1+0}+1} = \frac{4}{2} = 2.$$

1.8 Type 1.8: One-Sided Limit with Absolute Value

Problem 1.8: One-Sided Limit with $|x|$

Evaluate the one-sided limit:

$$\lim_{x \to 2^-} \frac{|x-2|}{x^2-4}$$

Gold-Standard Solution (Write your master solution template here)

Step 1. Apply the piecewise definition of the absolute value function:

$$|x-2| = \begin{cases} x-2, & \text{if } x \geq 2 \\ -(x-2), & \text{if } x < 2 \end{cases}$$

Since $x \to 2^-$, we have $x < 2$, which implies $|x-2| = -(x-2)$.

Step 2. Factor the denominator: $x^2 - 4 = (x-2)(x+2)$.

Step 3. Substitute and simplify the expression for $x < 2$:

$$\frac{|x-2|}{x^2-4} = \frac{-(x-2)}{(x-2)(x+2)} = -\frac{1}{x+2}.$$

Step 4. Evaluate the one-sided limit:

$$\lim_{x \to 2^-} \frac{|x-2|}{x^2-4} = \lim_{x \to 2^-} \left(-\frac{1}{x+2}\right) = -\frac{1}{2+2} = -\frac{1}{4}.$$

1.9 Type 1.9: L'Hôpital's Rule ($0/0$ or ∞/∞)

Problem 1.9: L'Hôpital's Rule

Evaluate the limit using L'Hôpital's Rule:

$$\lim_{x \to 0} \frac{e^{2x}-1}{\sin(3x)}$$

Gold-Standard Solution (Write your master solution template here)

Step 1. Verify the indeterminate form:

$$\lim_{x \to 0} (e^{2x}-1) = 1 - 1 = 0 \quad \text{and} \quad \lim_{x \to 0} \sin(3x) = 0 \implies \frac{0}{0} \quad \text{(Indeterminate Form).}$$

Step 2. Apply L'Hôpital's Rule by differentiating numerator and denominator:

$$\lim_{x \to 0} \frac{e^{2x}-1}{\sin(3x)} = \lim_{x \to 0} \frac{\frac{d}{dx}[e^{2x}-1]}{\frac{d}{dx}[\sin(3x)]}.$$

Step 3. Calculate the derivatives:

- $\frac{d}{dx}[e^{2x}-1] = 2e^{2x}$.

- $\frac{d}{dx}[\sin(3x)] = 3\cos(3x)$.

Step 4. Evaluate the resulting limit:

$$\lim_{x\to 0} \frac{2e^{2x}}{3\cos(3x)} = \frac{2e^0}{3\cos(0)} = \frac{2(1)}{3(1)} = \frac{2}{3}.$$

2 Basic Derivatives & Term-by-Term Differentiation

2.1 Type 2.1: Polynomial with Fractional and Negative Powers

Problem 2.1: Powers and Radicals

Given $f(x) = 3x^3 + \frac{3}{x} + 5\sqrt{x}$, find $f'(x)$ and evaluate $f'(1)$.

Gold-Standard Solution (Write your master solution template here)

Step 1. Rules and formula used in this problem are:

$$\begin{aligned} u^n &= \frac{1}{u^{-n}} \\ \sqrt[n]{u^m} &= u^{\frac{m}{n}} \\ [u^n]' &= nu^{n-1}u' \\ (k_1 f \pm k_2 g)' &= k_1 f' \pm k_2 g' \end{aligned}$$

Step 2. Find the derivative of f, we obtain

$$\begin{aligned} f'(x) &= \frac{d}{dx}\left(3x^3 + \frac{3}{x} + 5\sqrt{x}\right) \\ &= \frac{d}{dx}3x^3 + \frac{d}{dx}\frac{3}{x} + \frac{d}{dx}5\sqrt{x} \\ &= 3\frac{d}{dx}x^3 + 3\frac{d}{dx}x^{-1} + 5\frac{d}{dx}x^{\frac{1}{2}} \\ &= (3)(3)x^2 + (3)(-1)x^{-2} + (5)\left(\frac{1}{2}\right)x^{-\frac{1}{2}}. \end{aligned}$$

Thus,

$$f'(x) = 9x^2 - \frac{3}{x^2} + \frac{5}{2\sqrt{x}}$$

Step 3. Compute $f'(1)$ by putting $x = 1$ into $f'(x)$, then we have

$$\begin{aligned} f'(1) &= 9(1)^2 - \frac{3}{(1)^2} + \frac{5}{2\sqrt{1}} \\ &= 9(1) - \frac{3}{1} + \frac{5}{2} \\ &= 9 - 3 + \frac{5}{2} \\ &= \frac{17}{2}. \end{aligned}$$

Therefore, $f'(1) = \frac{17}{2}$.

2.2 Type 2.2: Basic Trigonometric Functions (Sine, Cosine, Secant, Tangent)

Problem 2.2: Trig Derivatives

Given $f(x) = 4\sin(3x) - 2\cos(3x)$, find $f'(x)$ and evaluate $f'(0)$.

Gold-Standard Solution (Write your master solution template here)

Step 1. Rules and formula used in this problem are:

$$
\begin{aligned}
[\sin(u)]' &= \cos(u)u' \\
[\cos(u)]' &= -\sin(u)u' \\
[u^n]' &= nu^{n-1}u' \\
(k_1 f \pm k_2 g)' &= k_1 f' \pm k_2 g'
\end{aligned}
$$

Step 2. Find the derivative of f, we obtain

$$
\begin{aligned}
f'(x) &= \frac{d}{dx}\left(4\sin(3x) - 2\cos(3x)\right) \\
&= \frac{d}{dx}4\sin(3x) - \frac{d}{dx}2\cos(3x) \\
&= 4\frac{d}{dx}\sin(3x) - 2\frac{d}{dx}\cos(3x) \\
&= 4\cos(3x)\frac{d}{dx}(3x) - 2(-\sin(3x))\frac{d}{dx}(3x) \\
&= (4)\cos(3x)(3) - (2)(-\sin(3x))(3).
\end{aligned}
$$

Thus,

$$f'(x) = 12\cos(3x) + 6\sin(3x)$$

Step 3. Compute $f'(0)$ by putting $x = 0$ into $f'(x)$, then we have

$$
\begin{aligned}
f'(0) &= 12\cos(3(0)) + 6\sin(3(0)) \\
&= 12\cos(0) + 6\sin(0) \\
&= 12(1) + 6(0) \\
&= 12.
\end{aligned}
$$

Therefore, $f'(0) = 12$.

2.3 Type 2.3: Natural and General Exponential Functions (e^{kx} and a^u)

Problem 2.3: Exponential Derivatives

Given $f(x) = 3^{2x+1} + 2e^{3x}$, find $f'(x)$ and evaluate $f'(0)$.

Gold-Standard Solution (Write your master solution template here)

Step 1. Rules and formula used in this problem are:

$$
\begin{aligned}
[a^u]' &= a^u\ln(a)u' \\
[e^u]' &= e^u u' \\
[u^n]' &= nu^{n-1}u' \\
[c]' &= 0 \\
(k_1 f \pm k_2 g)' &= k_1 f' \pm k_2 g'
\end{aligned}
$$

Step 2. Find the derivative of f, we obtain

$$\begin{aligned}
f'(x) &= \frac{d}{dx}\left(3^{2x+1} + 2e^{3x}\right)\\
&= \frac{d}{dx}3^{2x+1} + \frac{d}{dx}2e^{3x}\\
&= \frac{d}{dx}3^{2x+1} + 2\frac{d}{dx}e^{3x}\\
&= 3^{2x+1}\ln(3)\frac{d}{dx}(2x+1) + 2e^{3x}\frac{d}{dx}(3x)\\
&= 3^{2x+1}\ln(3)(2) + (2)e^{3x}(3).
\end{aligned}$$

Thus,

$$f'(x) = 2\ln(3)3^{2x+1} + 6e^{3x}$$

Step 3. Compute $f'(0)$ by putting $x = 0$ into $f'(x)$, then we have

$$\begin{aligned}
f'(0) &= 2\ln(3)3^{2(0)+1} + 6e^{3(0)}\\
&= 2\ln(3)3^{1} + 6e^{0}\\
&= 2\ln(3)(3) + 6(1)\\
&= 6\ln(3) + 6.
\end{aligned}$$

Therefore, $f'(0) = 6\ln(3) + 6$.

2.4 Type 2.4: Sum of Inverse Trigonometric Functions

Problem 2.4: Inverse Trig Derivatives

Given $f(x) = 3\arccos(4x) + 2\,\mathrm{arccot}(x)$, find $f'(x)$ and evaluate $f'(0)$.

Gold-Standard Solution (Write your master solution template here)

Step 1. Rules and formula used in this problem are:

$$\begin{aligned}
[\arccos(u)]' &= -\frac{1}{\sqrt{1-u^2}}u'\\
[\mathrm{arccot}(u)]' &= -\frac{1}{1+u^2}u'\\
(k_1 f \pm k_2 g)' &= k_1 f' \pm k_2 g'
\end{aligned}$$

Step 2. Find the derivative of f, we obtain

$$\begin{aligned} f'(x) &= \frac{d}{dx}\left(3\arccos(4x) + 2\,\mathrm{arccot}(x)\right) \\ &= \frac{d}{dx}3\arccos(4x) + \frac{d}{dx}2\,\mathrm{arccot}(x) \\ &= 3\frac{d}{dx}\arccos(4x) + 2\frac{d}{dx}\mathrm{arccot}(x) \\ &= 3\left(-\frac{1}{\sqrt{1-(4x)^2}}\right)\frac{d}{dx}(4x) + 2\left(-\frac{1}{1+x^2}\right)\frac{d}{dx}(x) \\ &= (3)\left(-\frac{1}{\sqrt{1-(4x)^2}}\right)(4) + (2)\left(-\frac{1}{1+x^2}\right)(1). \end{aligned}$$

Thus,

$$f'(x) = -\frac{12}{\sqrt{1-16x^2}} - \frac{2}{1+x^2}$$

Step 3. Compute $f'(0)$ by putting $x = 0$ into $f'(x)$, then we have

$$\begin{aligned} f'(0) &= -\frac{12}{\sqrt{1-16(0)^2}} - \frac{2}{1+(0)^2} \\ &= -\frac{12}{\sqrt{1-0}} - \frac{2}{1+0} \\ &= -\frac{12}{1} - \frac{2}{1} \\ &= -12 - 2 \\ &= -14. \end{aligned}$$

Therefore, $f'(0) = -14$.

3 Product Rule: $(uv)' = uv' + vu'$

3.1 Type 3.1: Polynomial × Exponential

Problem 3.1: Product Rule (Poly × Exp)

Given $f(x) = (2x+3)e^{2x}$, find $f'(x)$ and evaluate $f'(0)$.

Gold-Standard Solution (Write your master solution template here)

Step 1. Rules and formula used in this problem are:

$$
\begin{gathered}
(u \cdot v)' = uv' + vu' \\
[e^u]' = e^u u' \\
[u^n]' = nu^{n-1}u' \\
[c]' = 0 \\
(k_1 f \pm k_2 g)' = k_1 f' \pm k_2 g'
\end{gathered}
$$

Step 2. Find the derivative of f, we obtain

$$
\begin{aligned}
f'(x) &= \frac{d}{dx}(2x+3)e^{2x} \\
&= (2x+3)\frac{d}{dx}e^{2x} + e^{2x}\frac{d}{dx}(2x+3) \\
&= (2x+3)e^{2x}\frac{d}{dx}2x + e^{2x}\left[\frac{d}{dx}2x + \frac{d}{dx}3\right] \\
&= (2x+3)e^{2x}(2) + e^{2x}\left[2+0\right]
\end{aligned}
$$

Thus,

$$f'(x) = (4x+6)e^{2x} + 2e^{2x}.$$

Step 3. Compute $f'(0)$ by putting $x = 0$ into $f'(x)$, then we have

$$
\begin{aligned}
f'(0) &= (4(0)+6)e^{2(0)} + 2e^{2(0)} \\
&= (0+6)e^0 + 2e^0 \\
&= 6(1) + 2(1) \\
&= 6 + 2 \\
&= 8.
\end{aligned}
$$

Therefore, $f'(0) = 8$.

3.2 Type 3.2: Exponential × Trigonometric (Damped Oscillation)

Problem 3.2: Product Rule (Exp × Trig)

Given $f(x) = e^{-2x}\cos(3x)$, find $f'(x)$ and evaluate $f'(0)$.

Gold-Standard Solution (Write your master solution template here)

Step 1. Rules and formula used in this problem are:

$$
\begin{aligned}
(u \cdot v)' &= uv' + vu' \\
[e^u]' &= e^u u' \\
[\cos(u)]' &= -\sin(u)u' \\
[u^n]' &= nu^{n-1}u' \\
(k_1 f \pm k_2 g)' &= k_1 f' \pm k_2 g'
\end{aligned}
$$

Step 2. Apply the product rule:

$$f'(x) = \frac{d}{dx}\left[e^{-2x}\cos(3x)\right] = e^{-2x}\frac{d}{dx}\left[\cos(3x)\right] + \cos(3x)\frac{d}{dx}\left[e^{-2x}\right]. \tag{1}$$

Step 3. Differentiate each factor separately:

- Consider the derivative of $\cos(3x)$:

$$\frac{d}{dx}[\cos(3x)] = -\sin(3x)\frac{d}{dx}(3x) = -\sin(3x)\cdot 3\frac{d}{dx}x = -\sin(3x)\cdot 3(1) = -3\sin(3x).$$

- Consider the derivative of e^{-2x}:

$$\frac{d}{dx}\left[e^{-2x}\right] = e^{-2x}\frac{d}{dx}(-2x) = e^{-2x}\cdot(-2)\frac{d}{dx}x = e^{-2x}\cdot(-2)(1) = -2e^{-2x}.$$

Step 4. Combine these terms together, we obtain:

$$
\begin{aligned}
f'(x) &= e^{-2x}\frac{d}{dx}\left[\cos(3x)\right] + \cos(3x)\frac{d}{dx}\left[e^{-2x}\right] \\
&= e^{-2x}(-3\sin(3x)) + \cos(3x)(-2e^{-2x}) \\
&= -3e^{-2x}\sin(3x) - 2e^{-2x}\cos(3x) \\
&= -(3\sin(3x) + 2\cos(3x))e^{-2x}.
\end{aligned}
$$

Step 5. Evaluate at $x = 0$, since $e^0 = 1$, $\sin(0) = 0$, and $\cos(0) = 1$, we obtain:

$$f'(0) = -(3(0) + 2(1))e^0 = -2(1) = -2.$$

3.3 Type 3.3: Polynomial / Exponential × Inverse Trigonometric

Problem 3.3: Product Rule with Inverse Trig

Given $f(x) = (e^{3x} + 2x)\arctan(x)$, find $f'(x)$ and evaluate $f'(0)$.

Gold-Standard Solution (Write your master solution template here)

Step 1. Rules and formula used in this problem are:

$$(u \cdot v)' = uv' + vu'$$
$$[\arctan(u)]' = \frac{1}{1+u^2}u'$$
$$[e^u]' = e^u u'$$
$$[u^n]' = nu^{n-1}u'$$
$$(k_1 f \pm k_2 g)' = k_1 f' \pm k_2 g'$$

Step 2. Apply the product rule:

$$f'(x) = \frac{d}{dx}\left[(e^{3x}+2x)\arctan(x)\right] = (e^{3x}+2x)\frac{d}{dx}\left[\arctan(x)\right] + \arctan(x)\frac{d}{dx}\left[e^{3x}+2x\right]. \tag{2}$$

Step 3. Differentiate each factor separately:

- Consider the derivative of $\arctan(x)$:

$$\frac{d}{dx}[\arctan(x)] = \frac{1}{1+x^2}\frac{d}{dx}x = \frac{1}{1+x^2}(1) = \frac{1}{1+x^2}.$$

- Consider the derivative of $(e^{3x}+2x)$:

$$\frac{d}{dx}[e^{3x}+2x] = \frac{d}{dx}\left[e^{3x}\right] + 2\frac{d}{dx}x = e^{3x}\cdot 3(1) + 2(1) = 3e^{3x}+2.$$

Step 4. Combine these terms together, we obtain:

$$\begin{aligned} f'(x) &= (e^{3x}+2x)\frac{d}{dx}\left[\arctan(x)\right] + \arctan(x)\frac{d}{dx}\left[e^{3x}+2x\right] \\ &= (e^{3x}+2x)\left(\frac{1}{1+x^2}\right) + \arctan(x)(3e^{3x}+2) \\ &= \frac{e^{3x}+2x}{1+x^2} + (3e^{3x}+2)\arctan(x). \end{aligned}$$

Step 5. Evaluate at $x=0$, since $e^0 = 1$ and $\arctan(0) = 0$, we obtain:

$$f'(0) = \frac{1+0}{1+0} + (3(1)+2)\arctan(0) = 1 + 5(0) = 1.$$

4 Quotient Rule: $\left(\frac{u}{v}\right)' = \frac{vu'-uv'}{v^2}$

4.1 Type 4.1: Rational Polynomial Function

Problem 4.1: Quotient Rule (Polynomials)

Given $f(x) = \frac{2x+3}{x-4}$, find $f'(x)$ and evaluate $f'(0)$.

Gold-Standard Solution (Write your master solution template here)

Step 1. Rules and formula used in this problem are:

$$\left(\frac{u}{v}\right)' = \frac{vu' - uv'}{v^2}$$
$$[u^n]' = nu^{n-1}u'$$
$$[c]' = 0$$
$$(k_1 f \pm k_2 g)' = k_1 f' \pm k_2 g'$$

Step 2. Apply the quotient rule:

$$f'(x) = \frac{d}{dx}\left[\frac{2x+3}{x-4}\right] = \frac{(x-4)\frac{d}{dx}[2x+3] - (2x+3)\frac{d}{dx}[x-4]}{(x-4)^2}. \tag{3}$$

Step 3. Differentiate numerator and denominator separately:

- Consider the derivative of the numerator:

$$\frac{d}{dx}[2x+3] = 2\frac{d}{dx}x + \frac{d}{dx}(3) = 2(1) + 0 = 2.$$

- Consider the derivative of the denominator:

$$\frac{d}{dx}[x-4] = \frac{d}{dx}x - \frac{d}{dx}(4) = 1(1) - 0 = 1.$$

Step 4. Combine these terms together, we obtain:

$$\begin{aligned} f'(x) &= \frac{(x-4)(2) - (2x+3)(1)}{(x-4)^2} \\ &= \frac{2x - 8 - 2x - 3}{(x-4)^2} \\ &= -\frac{11}{(x-4)^2}. \end{aligned}$$

Step 5. Evaluate at $x = 0$, we obtain:

$$f'(0) = -\frac{11}{(0-4)^2} = -\frac{11}{16}.$$

4.2 Type 4.2: Trigonometric / Radical Quotient

Problem 4.2: Quotient Rule (Trig / Radical)

Given $f(x) = \frac{\sin(2x)}{\sqrt{x^2+1}}$, find $f'(x)$ and evaluate $f'(0)$.

Gold-Standard Solution (Write your master solution template here)

Step 1. Rules and formula used in this problem are:

$$\left(\frac{u}{v}\right)' = \frac{vu' - uv'}{v^2}$$
$$[\sin(u)]' = \cos(u)u'$$
$$[\sqrt{u}]' = \frac{1}{2\sqrt{u}}u'$$
$$[u^n]' = nu^{n-1}u'$$
$$[c]' = 0$$

Step 2. Apply the quotient rule:

$$f'(x) = \frac{d}{dx}\left[\frac{\sin(2x)}{\sqrt{x^2+1}}\right] = \frac{\sqrt{x^2+1}\,\frac{d}{dx}[\sin(2x)] - \sin(2x)\,\frac{d}{dx}\left[\sqrt{x^2+1}\right]}{\left(\sqrt{x^2+1}\right)^2}. \tag{4}$$

Step 3. Differentiate numerator and denominator separately:

- Consider the derivative of the numerator $\sin(2x)$:

$$\frac{d}{dx}[\sin(2x)] = \cos(2x)\frac{d}{dx}(2x) = \cos(2x)\cdot 2(1) = 2\cos(2x).$$

- Consider the derivative of the denominator $\sqrt{x^2+1}$:

$$\frac{d}{dx}\left[\sqrt{x^2+1}\right] = \frac{1}{2\sqrt{x^2+1}}\frac{d}{dx}[x^2+1] = \frac{2x(1)+0}{2\sqrt{x^2+1}} = \frac{x}{\sqrt{x^2+1}}.$$

Step 4. Combine these terms together, we obtain:

$$\begin{aligned} f'(x) &= \frac{\sqrt{x^2+1}(2\cos(2x)) - \sin(2x)\left(\frac{x}{\sqrt{x^2+1}}\right)}{x^2+1} \\ &= \frac{2(x^2+1)\cos(2x) - x\sin(2x)}{(x^2+1)^{\frac{3}{2}}}. \end{aligned}$$

Step 5. Evaluate at $x = 0$, since $\cos(0) = 1$ and $\sin(0) = 0$, we obtain:

$$f'(0) = \frac{2(0+1)(1) - 0(0)}{(0+1)^{\frac{3}{2}}} = \frac{2}{1} = 2.$$

5 Chain Rule & Composite Functions

5.1 Type 5.1: Radical with Quadratic Inside $\sqrt{u(x)}$

Problem 5.1: Chain Rule with Radical

Given $f(x) = \sqrt{2x^2 + 3x + 4}$, find $f'(x)$ and evaluate $f'(0)$.

Gold-Standard Solution (Write your master solution template here)

Step 1. Rules and formula used in this problem are:

$$\begin{aligned}
[\sqrt{u}]' &= \frac{1}{2\sqrt{u}}u' \\
[u^n]' &= nu^{n-1}u' \\
[c]' &= 0 \\
(k_1 f \pm k_2 g)' &= k_1 f' \pm k_2 g'
\end{aligned}$$

Step 2. Apply the chain rule by letting $u = 2x^2 + 3x + 4$:

$$f'(x) = \frac{d}{dx}\left[\sqrt{2x^2 + 3x + 4}\right] = \frac{1}{2\sqrt{2x^2 + 3x + 4}}\frac{d}{dx}\left[2x^2 + 3x + 4\right]. \tag{5}$$

Step 3. Differentiate the inner function:

$$\frac{d}{dx}[2x^2 + 3x + 4] = 2\frac{d}{dx}[x^2] + 3\frac{d}{dx}x + \frac{d}{dx}(4) = 2(2x \cdot 1) + 3(1) + 0 = 4x + 3.$$

Step 4. Combine these terms together, we obtain:

$$\begin{aligned}
f'(x) &= \frac{1}{2\sqrt{2x^2 + 3x + 4}}(4x + 3) \\
&= \frac{4x + 3}{2\sqrt{2x^2 + 3x + 4}}.
\end{aligned}$$

Step 5. Evaluate at $x = 0$, we obtain:

$$f'(0) = \frac{4(0) + 3}{2\sqrt{2(0)^2 + 3(0) + 4}} = \frac{3}{2\sqrt{4}} = \frac{3}{2(2)} = \frac{3}{4}.$$

5.2 Type 5.2: Logarithmic with Polynomial Inside $\ln(u(x))$

Problem 5.2: Chain Rule with Natural Log

Given $f(x) = \ln(x^2 + 3x + 2)$, find $f'(x)$ and evaluate $f'(0)$.

Gold-Standard Solution (Write your master solution template here)

Step 1. Rules and formula used in this problem are:

$$[\ln(u)]' = \frac{1}{u}u'$$
$$[u^n]' = nu^{n-1}u'$$
$$[c]' = 0$$
$$(k_1 f \pm k_2 g)' = k_1 f' \pm k_2 g'$$

Step 2. Apply the chain rule by letting $u = x^2 + 3x + 2$:

$$f'(x) = \frac{d}{dx}\left[\ln(x^2 + 3x + 2)\right] = \frac{1}{x^2 + 3x + 2}\frac{d}{dx}\left[x^2 + 3x + 2\right]. \tag{6}$$

Step 3. Differentiate the inner function:

$$\frac{d}{dx}[x^2 + 3x + 2] = \frac{d}{dx}[x^2] + 3\frac{d}{dx}x + \frac{d}{dx}(2) = 2x(1) + 3(1) + 0 = 2x + 3.$$

Step 4. Combine these terms together, we obtain:

$$f'(x) = \frac{2x + 3}{x^2 + 3x + 2}.$$

Step 5. Evaluate at $x = 0$, we obtain:

$$f'(0) = \frac{2(0) + 3}{0^2 + 3(0) + 2} = \frac{3}{2}.$$

5.3 Type 5.3: Trigonometric with Linear / Quadratic Inside

Problem 5.3: Chain Rule with Cotangent

Given $f(x) = 2\cot\left(3x + \frac{\pi}{4}\right)$, find $f'(x)$ and evaluate $f'(0)$.

Gold-Standard Solution (Write your master solution template here)

Step 1. Rules and formula used in this problem are:

$$[\cot(u)]' = -\csc^2(u)u'$$
$$[u^n]' = nu^{n-1}u'$$
$$[c]' = 0$$
$$(k \cdot g)' = k \cdot g'$$

Step 2. Apply constant multiple and chain rule by letting $u = 3x + \frac{\pi}{4}$:

$$f'(x) = 2\frac{d}{dx}\left[\cot\left(3x + \frac{\pi}{4}\right)\right] = 2\left(-\csc^2\left(3x + \frac{\pi}{4}\right)\frac{d}{dx}\left[3x + \frac{\pi}{4}\right]\right). \tag{7}$$

Step 3. Differentiate the inner function:

$$\frac{d}{dx}\left[3x + \frac{\pi}{4}\right] = 3\frac{d}{dx}x + \frac{d}{dx}\left(\frac{\pi}{4}\right) = 3(1) + 0 = 3.$$

Step 4. Combine these terms together, we obtain:

$$\begin{aligned} f'(x) &= 2\left(-\csc^2\left(3x+\frac{\pi}{4}\right)\cdot 3\right) \\ &= -6\csc^2\left(3x+\frac{\pi}{4}\right). \end{aligned}$$

Step 5. Evaluate at $x = 0$, since $\csc\left(\frac{\pi}{4}\right) = \sqrt{2}$, we obtain $\csc^2\left(\frac{\pi}{4}\right) = 2$:

$$f'(0) = -6\csc^2\left(\frac{\pi}{4}\right) = -6(2) = -12.$$

5.4 Type 5.4: Inverse Trigonometric with Chain Rule

Problem 5.4: Chain Rule with ArcSec

Given $f(x) = 2\,\text{arcsec}(3x)$, find $f'(x)$ and evaluate $f'(1)$.

Gold-Standard Solution (Write your master solution template here)

Step 1. Rules and formula used in this problem are:

$$\begin{aligned} [\text{arcsec}(u)]' &= \frac{1}{|u|\sqrt{u^2-1}}u' \\ [u^n]' &= nu^{n-1}u' \\ (k\cdot g)' &= k\cdot g' \end{aligned}$$

Step 2. Apply constant multiple and chain rule by letting $u = 3x$:

$$f'(x) = 2\frac{d}{dx}\left[\text{arcsec}(3x)\right] = 2\left(\frac{1}{|3x|\sqrt{(3x)^2-1}}\frac{d}{dx}[3x]\right). \tag{8}$$

Step 3. Differentiate the inner function:

$$\frac{d}{dx}[3x] = 3\frac{d}{dx}x = 3(1) = 3.$$

Step 4. Combine these terms together, we obtain:

$$\begin{aligned} f'(x) &= 2\left(\frac{1}{|3x|\sqrt{9x^2-1}}\cdot 3\right) \\ &= \frac{6}{|3x|\sqrt{9x^2-1}} = \frac{2}{|x|\sqrt{9x^2-1}}. \end{aligned}$$

Step 5. Evaluate at $x = 1$, we obtain:

$$f'(1) = \frac{2}{|1|\sqrt{9(1)^2-1}} = \frac{2}{\sqrt{8}} = \frac{2}{2\sqrt{2}} = \frac{\sqrt{2}}{2}.$$

5.5 Type 5.5: Nested Composite Function (Trig(Inverse Trig(kx)))

Problem 5.5: Nested Trig of Inverse Trig

Given $f(x) = \tan(\arcsin(3x))$, find $f'(x)$ and evaluate $f'(0)$.

Gold-Standard Solution (Write your master solution template here)

Step 1. Rules and formula used in this problem are:

$$[\tan(u)]' = \sec^2(u)u'$$
$$[\arcsin(u)]' = \frac{1}{\sqrt{1-u^2}}u'$$
$$[u^n]' = nu^{n-1}u'$$

Step 2. Apply the chain rule from outer to inner functions:

$$f'(x) = \frac{d}{dx}\left[\tan(\arcsin(3x))\right] = \sec^2(\arcsin(3x))\frac{d}{dx}\left[\arcsin(3x)\right]. \tag{9}$$

Step 3. Differentiate the inner function $\arcsin(3x)$:

$$\frac{d}{dx}[\arcsin(3x)] = \frac{1}{\sqrt{1-(3x)^2}}\frac{d}{dx}[3x] = \frac{1}{\sqrt{1-9x^2}} \cdot 3\frac{d}{dx}x = \frac{3}{\sqrt{1-9x^2}}.$$

Step 4. Combine these terms together, we obtain:

$$f'(x) = \sec^2(\arcsin(3x)) \cdot \frac{3}{\sqrt{1-9x^2}} = \frac{3\sec^2(\arcsin(3x))}{\sqrt{1-9x^2}}.$$

Step 5. Evaluate at $x = 0$, since $\arcsin(0) = 0$ and $\sec(0) = 1$, we obtain:

$$f'(0) = \frac{3\sec^2(0)}{\sqrt{1-0}} = \frac{3(1)^2}{1} = 3.$$

5.6 Type 5.6: Logarithmic Differentiation ($y = x^{kx}$)

Problem 5.6: Logarithmic Differentiation

Given $f(x) = x^{3x}$ for $x > 0$, find $f'(x)$ and evaluate $f'(1)$.

Gold-Standard Solution (Write your master solution template here)

Step 1. Rules and properties used in logarithmic differentiation:

$$\ln(u^r) = r\ln(u)$$
$$[\ln(y)]' = \frac{1}{y}y'$$
$$(u \cdot v)' = uv' + vu'$$
$$[\ln(x)]' = \frac{1}{x}$$

Step 2. Take natural logarithm of both sides and differentiate: Let $y = x^{3x}$. Then $\ln(y) = 3x\ln(x)$.

$$\frac{d}{dx}[\ln(y)] = \frac{d}{dx}[3x\ln(x)]$$
$$\frac{1}{y}\frac{dy}{dx} = 3x\frac{d}{dx}[\ln(x)] + \ln(x)\frac{d}{dx}[3x].$$

Step 3. Differentiate each part on the right-hand side:

- $3x\frac{d}{dx}[\ln(x)] = 3x\left(\frac{1}{x}\right) = 3.$
- $\ln(x)\frac{d}{dx}[3x] = \ln(x)\cdot 3(1) = 3\ln(x).$

Step 4. Combine and solve for $\frac{dy}{dx} = f'(x)$:

$$\frac{1}{y}\frac{dy}{dx} = 3 + 3\ln(x) = 3(1+\ln(x))$$
$$f'(x) = y\cdot 3(1+\ln(x))$$
$$f'(x) = 3x^{3x}(1+\ln(x)).$$

Step 5. Evaluate at $x = 1$, since $\ln(1) = 0$ and $1^3 = 1$, we obtain:

$$f'(1) = 3(1)^3(1+\ln(1)) = 3(1)(1+0) = 3.$$

5.7 Type 5.7: Higher-Order Derivative ($f''(x)$)

Problem 5.7: Second Derivative

Given $f(x) = x^2e^{2x}$, find $f''(x)$ and evaluate $f''(0)$.

Gold-Standard Solution (Write your master solution template here)

Step 1. Rules and formula used in this problem are:

$$(u\cdot v)' = uv' + vu'$$
$$[e^u]' = e^u u'$$
$$[u^n]' = nu^{n-1}u'$$
$$(k_1 f \pm k_2 g)' = k_1 f' \pm k_2 g'$$

Step 2. Find the first derivative $f'(x)$ using the product rule:

$$f'(x) = \frac{d}{dx}\left[x^2e^{2x}\right] = x^2\frac{d}{dx}[e^{2x}] + e^{2x}\frac{d}{dx}[x^2]. \tag{10}$$

Step 3. Differentiate each component:

$$\frac{d}{dx}[e^{2x}] = e^{2x}\frac{d}{dx}[2x] = 2e^{2x}$$
$$\frac{d}{dx}[x^2] = 2x\frac{dx}{dx} = 2x(1) = 2x$$

Step 4. Combine and simplify $f'(x)$:

$$f'(x) = x^2(2e^{2x}) + e^{2x}(2x) = 2x^2e^{2x} + 2xe^{2x} = (2x^2 + 2x)e^{2x}.$$

Step 5. Find the second derivative $f''(x)$ by differentiating $f'(x)$ again:

$$\begin{aligned}
f''(x) &= \frac{d}{dx}\left[(2x^2 + 2x)e^{2x}\right] \\
&= (2x^2 + 2x)\frac{d}{dx}[e^{2x}] + e^{2x}\frac{d}{dx}[2x^2 + 2x] \\
&= (2x^2 + 2x)(2e^{2x}) + e^{2x}(4x + 2) \\
&= (4x^2 + 4x + 4x + 2)e^{2x} \\
&= (4x^2 + 8x + 2)e^{2x}.
\end{aligned}$$

Evaluate at $x = 0$:

$$f''(0) = (4(0)^2 + 8(0) + 2)e^0 = 2(1) = 2.$$

6 Implicit Differentiation

6.1 Type 6.1: Circle Equation ($x^2 + y^2 = r^2$)

Problem 6.1: Implicit Differentiation of a Circle

Find $\frac{dy}{dx}$ for the equation $x^2 + y^2 = 25$ at the point $(3, 4)$.

Gold-Standard Solution (Write your master solution template here)

Step 1. Rules and formula used in this problem are:

$$\frac{d}{dx}[x^n] = nx^{n-1}$$
$$\frac{d}{dx}[y^n] = ny^{n-1}\frac{dy}{dx}$$
$$\frac{d}{dx}[c] = 0$$

Step 2. Differentiate both sides of the equation with respect to x:

$$\frac{d}{dx}\left[x^2 + y^2\right] = \frac{d}{dx}[25]$$
$$\frac{d}{dx}[x^2] + \frac{d}{dx}[y^2] = 0.$$

Step 3. Calculate each derivative term:

$$\frac{d}{dx}[x^2] = 2x\frac{dx}{dx} = 2x(1) = 2x$$
$$\frac{d}{dx}[y^2] = 2y\frac{dy}{dx}$$

Step 4. Substitute back and isolate $\frac{dy}{dx}$:

$$2x + 2y\frac{dy}{dx} = 0$$
$$2y\frac{dy}{dx} = -2x$$
$$\frac{dy}{dx} = -\frac{2x}{2y} = -\frac{x}{y}.$$

Step 5. Substitute the given point $(x, y) = (3, 4)$:

$$\left.\frac{dy}{dx}\right|_{(3,4)} = -\frac{3}{4}.$$

6.2 Type 6.2: Mixed Polynomial with xy Product Term

Problem 6.2: Implicit Differentiation with Product Rule

Find $\frac{dy}{dx}$ for the equation $x^3 + y^3 - 3xy = 5$ at the point (x_0, y_0).

Gold-Standard Solution (Write your master solution template here)

Step 1. Rules and formula used in this problem are:

$$\begin{aligned}\frac{d}{dx}[u \cdot v] &= u\frac{dv}{dx} + v\frac{du}{dx}\\ \frac{d}{dx}[y^n] &= ny^{n-1}\frac{dy}{dx}\\ \frac{d}{dx}[x^n] &= nx^{n-1}\\ \frac{d}{dx}[c] &= 0\end{aligned}$$

Step 2. Differentiate both sides of the equation with respect to x:

$$\begin{aligned}\frac{d}{dx}\left[x^3 + y^3 - 3xy\right] &= \frac{d}{dx}[5]\\ \frac{d}{dx}[x^3] + \frac{d}{dx}[y^3] - 3\frac{d}{dx}[xy] &= 0.\end{aligned}$$

Step 3. Calculate each derivative term:

$$\begin{aligned}&\frac{d}{dx}[x^3] = 3x^2\frac{dx}{dx} = 3x^2(1) = 3x^2\\ &\frac{d}{dx}[y^3] = 3y^2\frac{dy}{dx}\\ &\frac{d}{dx}[xy] = x\frac{dy}{dx} + y\frac{dx}{dx} = x\frac{dy}{dx} + y(1) = x\frac{dy}{dx} + y\end{aligned}$$

Step 4. Combine and isolate $\frac{dy}{dx}$:

$$\begin{aligned}3x^2 + 3y^2\frac{dy}{dx} - 3\left(x\frac{dy}{dx} + y\right) &= 0\\ 3x^2 + 3y^2\frac{dy}{dx} - 3x\frac{dy}{dx} - 3y &= 0\\ (3y^2 - 3x)\frac{dy}{dx} &= 3y - 3x^2\\ \frac{dy}{dx} &= \frac{3y - 3x^2}{3y^2 - 3x} = \frac{y - x^2}{y^2 - x}.\end{aligned}$$

Step 5. Evaluate at the given point (x_0, y_0), we obtain:

$$\left.\frac{dy}{dx}\right|_{(x_0,y_0)} = \frac{y_0 - x_0^2}{y_0^2 - x_0}.$$

7 Tangent & Normal Lines

7.1 Type 7.1: Equation of the Tangent Line

Problem 7.1: Tangent Line Equation

Find the slope of the tangent line m_t and the equation of the tangent line to the curve $y = x^2 - 3x + 2$ at $x = 2$.

Gold-Standard Solution (Write your master solution template here)

Step 1. Formulas for the tangent line:

$$\begin{aligned} m_t &= f'(x_0) \\ y - y_0 &= m_t(x - x_0) \end{aligned}$$

Step 2. Find the derivative $f'(x)$:

$$f'(x) = \frac{d}{dx}\left[x^2 - 3x + 2\right] = 2x - 3. \tag{11}$$

Step 3. Find the point of tangency (x_0, y_0) by evaluating $y(2)$:

$$y_0 = f(2) = (2)^2 - 3(2) + 2 = 4 - 6 + 2 = 0$$

implies $(x_0, y_0) = (2, 0)$.

Step 4. Find the slope of the tangent line m_t:

$$m_t = f'(2) = 2(2) - 3 = 4 - 3 = 1.$$

Step 5. Construct the equation of the tangent line:

$$\begin{aligned} y - y_0 &= m_t(x - x_0) \\ y - 0 &= 1(x - 2) \\ y &= x - 2 \quad (\text{or } x - y - 2 = 0)\,. \end{aligned}$$

7.2 Type 7.2: Equation of the Normal Line

Problem 7.2: Normal Line Equation

Find the slope of the normal line m_n and the equation of the normal line to the curve $y = x^2 - 3x + 2$ at $x = 2$.

Gold-Standard Solution (Write your master solution template here)

Step 1. Formulas for the normal line:

$$\begin{aligned} m_t &= f'(x_0) \\ m_n &= -\frac{1}{m_t} \quad (m_t \neq 0) \\ y - y_0 &= m_n(x - x_0) \end{aligned}$$

Step 2. Find the derivative $f'(x)$:

$$f'(x) = \frac{d}{dx}\left[x^2 - 3x + 2\right] = 2x - 3. \tag{12}$$

Step 3. Find the point (x_0, y_0) by evaluating $y(2)$:

$$y_0 = f(2) = (2)^2 - 3(2) + 2 = 4 - 6 + 2 = 0$$

implies $(x_0, y_0) = (2, 0)$.

Step 4. Find the slope of the normal line m_n:

$$\begin{aligned} m_t &= f'(2) = 2(2) - 3 = 1 \\ m_n &= -\frac{1}{m_t} = -\frac{1}{1} = -1. \end{aligned}$$

Step 5. Construct the equation of the normal line:

$$\begin{aligned} y - y_0 &= m_n(x - x_0) \\ y - 0 &= -1(x - 2) \\ y &= -x + 2 \quad (\text{or } x + y - 2 = 0)\,. \end{aligned}$$